

Given reconstructed survival IPD from published KM curves and baseline summaries, how can we construct plausible patient-level covariate matrices that satisfy reporter marginal and subgroup constraints and quantify how uncertainty in these reconstructions affect downstream analysis?

Proposed Pipeline:

1. *Constraint Extraction:* For each trial, collect whatever is reported, like marginal counts, continuous summaries (arm-specific and subgroup if available) and known clinical/logical constraints, and organize them into hard and soft constraints. This acts as a “constraint library”.
2. *Generate feasible marginal covariates:* For categorical variables, we can use exact counts. For continuous variables like age, we can utilize distributional generation matching available summaries:
 - a. Mean and SD use truncated or skew normal
 - b. Median and IQR use quantile-matched distribution
 - c. Median only use weakly informative distribution with median centering
3. *Jointly Assign Covariates:* Some possible (sketch) methods:
 - a. Generate each marginal exactly then repeatedly swap labels to improve plausibility
 - b. If we have marginal tables and partial joint tables, we can try to estimate a joint contingency table
 - c. Integer programming?
 - d. Simulated Annealing over covariate matrices
 - e. Assign covariate profiles to reconstructed survival records while trying to preserve risk orderine
 - f. I think a possible first version is to do exact marginal matching with MCMC swaps and add soft penalties for clinical dependence
4. *Survival Based Assignment:* Right now the reconstruction is independent of survival outcomes, so we could make a plausibility score: $L = \text{marginal} + \text{joint} + \text{survival}$ where survival part rewards assignments where high risk covariates are more common among earlier events
5. *Ensemble:* For each study, generate multiple reconstructions that satisfy published constraints but differ in joint assignment, and combine with RESOLVE so uncertainty can be assessed
6. *Propagating uncertainty downstream:* We can propagate this uncertainty through the two ways proposed by Chelsea:
 - a. Adjusted treatment effect
 - b. Treatment-effect heterogeneity (more useful but more complex, since if the covariates are reconstructed, treatment-effect interactions will be very sensitive. That sensitivity is interesting, but it could also be messy)

Evaluation Strategy

It would be helpful to design a simulation benchmark. We generate true full IPD and then hide the design matrix, only keeping a variety of summaries, KM curves, and risk tables. We can evaluate in two aspects:

1. Covariates: Marginal error, pairwise correlation error, plausibility score
2. Downtream effect: bias of adjusted treatment effect, coverage, effect modifier accuracy
3. Survival-informed coveriate assignment can be assed through survival agreement if subgroup curves are available

Edge cases

We can test settings with missing subgroup summaries, strong covariate dependence, censoring heavy tails, and delayed treatment effect

Discussion:

- MAPLE-esque methods can fit in the joint label reconstruction step
- Algorithm sketch

1. Initialize Design by exact marginal matching
2. Define loss as a weighted sum of marginals, pairwise, clinical, and survival
3. Propose swaps
 - a. Swap covariate labels between two patients
 - b. Swap joint profiles
 - c. Perturb continuous age values
4. Accept/Reject through simulated annealing
5. Save multiple low-loss samples after burn-in